



Regulation: R22

Code No: 22PY104/1

I B. Tech I Semester Regular Examinations – January, 2023

PHYSICS FOR ELECTRONIC ENGINEERS

Time: 150 Min

(ECE)

Max. Marks: 80M

SECTION – A

Answer all Four questions

4×10M=40M

1. a) Relate lattice parameter and atomic radius in the case of BCC structure. [2M]
 b) Face-centered cubic structure is closely packed when compared to body-centered cubic structure. Justify. [4M]
 c) The metal Rhodium has FCC crystal structure. If the angle of diffraction for the (311) set of planes occurs at 36.12° (first-order reflection) when monochromatic X-radiation having a wavelength 0.0711nm is used, compute the following. (i) The inter-planar spacing for this set of planes. (ii) The atomic radius for Rh atom. [4M]

2. a) The Schroedinger's one-dimensional wave equation for a particle is $\frac{d^2\psi}{dx^2} + \frac{2m(E - V)}{\hbar^2} \psi = 0$ and its corresponding wave function is $\psi(x) = A \sin \frac{n\pi x}{a}$. Rewrite the above expressions for a free particle in space. [2M]

- b) Discuss the effect of temperature on Fermi-Dirac distribution of free electrons in a metal. [4M]
 c) Find the spacing between the lowest two energy levels of an electron confined to a potential box of length 0.2 nm. [4M]

3. a) Deduce Coulomb's law from Gauss law. [4M]
 b) An isolated charged sphere of charge Q has radius R. Deduce an expression for the electric field at a distance r from the center of the sphere ($r > R$) and on the sphere ($r=R$) using Gauss law. [6M]

4. a) Discuss V-I characteristics of a solar cell and demonstrate the relationship between fill-factor and efficiency of a solar cell. [4M]
 b) Distinguish between PIN diode and Avalanche photo diode. [2M]
 c) The fill-factor of a solar cell is 0.5, open circuit-voltage is 0.55V when exposed to light of an input power of 500 mW. The cell area is 2cm^2 . Calculate its conversion efficiency. Given short-circuit current is 200 mA. [4M]

SECTION - B

Answer all Two questions

2×20M=40M

5. a) The inter-planar spacing d_{hkl} for planes in a unit cell having orthorhombic geometry is given by $\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$ where a, b and c are the lattice parameters. Modify this expression for crystals having cubic and tetragonal symmetries. [6M]
- b) Discuss the formulation of diffraction pattern on the photographic film in powder X-ray diffraction and how will you obtain glancing angle? [6M]
- c) The wave function for a particle in n^{th} state in an infinite potential box of width “a” is $\psi(x) = A \sin \frac{n\pi x}{a}$. Determine the constant A using the condition for normalized wave function. [4M]
- d) Sketch energy levels and wave functions for the levels $n=2,3,4$ for a particle in a one dimensional rigid box. [4M]
6. a) Distinguish between spontaneous emission and stimulated emission. [4M]
- b) Interpret the lasing action in diode laser from the prospective of essential components required for it. [6M]
- c) Differentiate Biot-Savart’ law from Ampere’s law and outline the expression for magnetic field as per Biot-Savart’ law. [5M]
- d) Estimate the magnetic field using Biot-Savart’s law on the axis of a circular coil of radius 0.2m, with 100 turns and carrying a steady current 0.5 Amp at a point P which is at a distance 0.15m from the centre of the coil. (Given , $\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$) [5M]